AGILE SPRINT PLANNING & USER STORY WORKSHOP

Smart Renewable Battery Recycling Platform

1. Introduction

The Smart Renewable Battery Recycling Platform is developed to improve the process of collecting and recycling used batteries. Nowadays, batteries are widely used in electric vehicles, mobile phones, laptops, and renewable energy systems. Improper disposal of batteries can cause environmental pollution and health hazards. This platform helps recycling industries monitor battery inventory, manage recycling activities, predict hazardous battery conditions, track transportation, generate sustainability reports, and notify recycling operators. By using this system, recycling becomes safer, faster, and more efficient.

2. Project Vision

The main vision of this project is to build a smart software platform that supports safe and efficient battery recycling. The system should help industries monitor battery stock, reduce manual work, improve safety during recycling, and provide useful reports for environmental sustainability. The platform also aims to reduce battery waste and encourage responsible recycling practices.

3. Project Objectives

The objectives of this project are:

- To monitor battery inventory.

- To optimize recycling operations.
- To predict hazardous battery conditions.
- To track battery transportation.
- To generate sustainability reports.
- To notify recycling operators about important events.

4. Stakeholders

The success of this project depends on different stakeholders who use or manage the system.

- **System Administrator** – Manages users, system security, and maintenance.
- **Plant Manager** – Monitors recycling operations and assigns work.
- **Recycling Operator** – Updates recycling activities and receives notifications.
- **Safety Officer** – Monitors hazardous battery conditions.
- **Logistics Manager** – Tracks battery transportation.
- **Environmental Officer** – Reviews sustainability reports.
- **Battery Collection Centers** – Send batteries for recycling.
- **Government Authorities** – Ensure environmental regulations are followed.

5. Epics and User Stories

Epic 1 – Battery Inventory Management

User Story

As a recycling operator, I want to register newly collected batteries so that the inventory remains updated.

User Story

As a plant manager, I want to view battery inventory so that I can plan recycling operations effectively.

Epic 2 – Recycling Operations

User Story

As a plant manager, I want to assign recycling tasks to operators so that work is completed efficiently.

User Story

As a recycling operator, I want to update the status of assigned tasks so that the manager can monitor progress.

Epic 3 – Hazard Detection

User Story

As a safety officer, I want the system to identify hazardous battery conditions so that accidents can be prevented.

User Story

As a recycling operator, I want to receive emergency alerts so that I can take immediate action.

Epic 4 – Logistics Tracking

User Story

As a logistics manager, I want to monitor battery transportation so that I know the current delivery status.

Epic 5 – Sustainability Reporting

User Story

As an environmental officer, I want the system to generate sustainability reports so that environmental performance can be evaluated.

6. Acceptance Criteria

1. **Battery Inventory**

Given the operator has logged into the system,

When a new battery is added,

Then the battery details should be stored successfully in the inventory.

2.Recycling Task Management

Given recycling tasks are available,

When the manager assigns a task,

Then the assigned operator should receive the task information immediately.

3.Hazard Detection

Given battery monitoring data is available,

When abnormal temperature or voltage is detected,

Then the system should generate a warning notification.

4.Logistics Tracking

Given a battery shipment has started,

When transportation details are updated,

Then the current shipment status should be displayed.

5.Sustainability Report

Given recycling information is available,

When the report is generated,

Then the system should display the sustainability report successfully.

7. Product Backlog Prioritization

The Product Backlog is a list of all the features that need to be developed for the Smart Renewable Battery Recycling Platform. These features are prioritized based on business value, user requirements, and project dependencies. The development team focuses on implementing the most important features first so that the system delivers value at every stage of development.

- The highest priority is given to the **User Login and Authentication** module because users must securely access the system before using any other feature.
- The next priority is **Battery Inventory Management**, as it is the core functionality of the platform. This module allows operators to record, update, and monitor battery information.
- After inventory management, the **Recycling Task Management** module is developed. This feature enables the plant manager to assign recycling tasks and allows operators to update the progress of their assigned work.
- The **Hazard Detection and Alert System** is also considered a high-priority feature because it helps identify dangerous battery conditions and immediately notifies operators to prevent accidents.
- Once the core modules are completed, the team focuses on **Logistics Tracking**, which helps monitor battery transportation between collection centers and recycling plants.
- The **Notification System** is implemented to keep users informed about task assignments, shipment updates, and emergency alerts.
- Finally, the **Sustainability Report Generation** module is developed. This feature uses data collected from the other modules to generate reports showing recycling performance and environmental impact.
- By following this order, the development team can deliver the most essential features first while ensuring smooth project development.

8. Sprint Backlog

- The first sprint is planned for a duration of **two weeks**. During this sprint, the development team focuses on implementing the most important features required for the initial version of the application.
- The sprint begins with the development of the **User Login Module**, where the login page is designed and user authentication is implemented. This ensures that only authorized users can access the system.
- The team then develops the **Battery Inventory Management Module**. This includes creating the database structure, designing inventory forms, and implementing functions to add, edit, and view battery records.
- Next, the **Recycling Task Management Module** is developed. The plant manager will be able to assign tasks to recycling operators, and operators can update the status of their work after completing each task.
- The final module included in Sprint 1 is the **Hazard Detection System**. This module is responsible for identifying abnormal battery conditions and sending warning notifications whenever a potential risk is detected.
- After completing these modules, the development team performs unit testing to verify that each feature works correctly before presenting the sprint outcome during the sprint review meeting
- Story Point Estimation is used to measure the amount of effort required to complete each user story. In this project, the development team follows the **Fibonacci sequence (1, 2, 3, 5, 8, and 13)** because it provides a simple way to estimate work based on complexity and uncertainty.
- The **User Login Module** is assigned **3 story points** because it involves basic authentication and simple validation.
- The **Battery Inventory Management Module** is estimated at **5 story points** since it requires database operations, inventory updates, and data validation.

- The **Recycling Task Management Module** is assigned **8 story points** because it includes task assignment, task tracking, and interaction between managers and operators.
- The **Hazard Detection Module** receives **13 story points** because it is the most complex part of the system. It requires monitoring battery conditions, analyzing data, and generating emergency alerts.
- The **Logistics Tracking Module** is estimated at **8 story points** because it involves tracking battery transportation and updating shipment information.
- The **Notification System** is assigned **5 story points** since it requires sending alerts and updates to different users.
- The **Sustainability Report Module** is estimated at **3 story points** because it mainly generates reports using existing system data.
- Using the Fibonacci estimation method helps the development team understand the complexity of each feature and plan the sprint more effectively.

10. Sprint Goal

The main goal of Sprint 1 is to develop the core functionalities of the Smart Renewable Battery Recycling Platform. At the end of the sprint, users should be able to log into the system, manage battery inventory, assign recycling tasks, and receive hazard alerts. Completing these modules provides a strong foundation for implementing the remaining features in future sprints.

11. Expected Sprint Deliverables

At the end of Sprint 1, the following deliverables are expected:

- A secure User Login Module.
- A fully functional Battery Inventory Management Module.
- A Recycling Task Management Module for assigning and updating tasks.
- A Hazard Detection Module with alert notifications.
- A basic user interface for system interaction.
- A database to store battery and user information.
- Unit testing results for all completed modules.
- A working prototype that can be demonstrated during the sprint review.

12. Conclusion

The Smart Renewable Battery Recycling Platform is designed to improve the battery recycling process by using Agile software development practices. During Sprint Planning, the project requirements were divided into epics, user stories, and development tasks. The product backlog was prioritized based on business value, and story points were estimated using the Fibonacci sequence. This approach helps the development team focus on important features first while maintaining flexibility for future improvements. Overall, Agile Sprint Planning provides a clear roadmap for developing a reliable, efficient, and environmentally friendly battery recycling platform.

13. References

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